



BLUETOOTH TECHNOLOGY

In the article 'Understanding Reliability in Bluetooth Technology' published in April issue, in the last line on page 44 it is written that "Bluetooth LE divides the 2.4GHz band into 40 channels, each 2MHz wide." Should it not be 60MHz mathematically instead of 2MHz? Please clarify this statement technically.

Sachin Kumar

The author Martin Woolley replies:

The current wording is correct and does not require clarification. Please refer the Bluetooth Core Specification Version 5.2,

Twike after taking over the rights of the SwissLEM AG in 2002. I am sorry for the inadvertent mistake!

...

GPS ANALOGUE CLOCK

Thank you very much for publishing the exciting project titled 'Try Making This TFT-Based GPS Analogue Clock Using Arduino' in April 2020 issue. I have the clock working but am facing a small issue with the hours and minutes hands that blink every second. Do you think it is due to my TFT shield, or is there a way to refresh the TFT faster with coding?

And is there a way to update the daylight saving time (DST) automatically? I would like to have UTC + 10 hours on the first Sunday of April and UTC + 11 on the first Sunday of October. I am new to Arduino. If you can help me add automatic DST to this clock, that will be great.

Sam

Melbourne, Australia

objective will be achieved. The 'if' statement will work, but how long is not defined. Therefore, one more 'if' statement should be used within the first 'if' statement, if(mm < 10) hh = hh + 10; the same for the next correction to be taken care of.

...

CONTACTLESS SWITCH FOR SMART HOME

In the 'Gesture-Controlled Contactless Switch For Smart Home' DIY project published in December 2020 issue, the Arduino Pro Micro, Nano, and Mini boards are mentioned. I have bought Arduino Nano board but when I checked on Google, it is showing that Arduino Nano won't work with it. Please help.

Sathvi Gandu

☐ In 'Gesture-Controlled Contactless Switch For Smart Home' DIY project which Arduino board is used for programming?

Deny

EFY. The project will work with any of the three Arduino boards: Micro, Nano, and Mini. The circuit diagram shown is for Arduino Pro Micro. However, if the Arduino Nano version 3.0 is used, pin A4 is for serial data line (SDA) and pin A5 is for serial clock line (SCL) for I2C communications. Circuit and code should be changed accordingly. USB connector is missing in Arduino Pro Mini board. For programming or uploading the code to Arduino Pro Mini, you need a USB-to-TTL adaptor or SPI port. Here too, pin A4 is SDA and pin A5 is SCL for I2C communications.

Note. Most articles published in EFY magazine can also be read at our website <https://www.electronicshobby.com> after some time, including DIY articles whose components can be purchased from our [kitsnspares.com](https://www.kitsnspares.com) website (or by writing to info@kitsnspares.com).

BLUETOOTH CORE SPECIFICATION Version 5.2 | Vol 6, Part A page 2831

Physical Layer Specification

2 FREQUENCY BANDS AND CHANNEL ARRANGEMENT

The LE system operates in the 2.4 GHz ISM band at 2400-2483.5 MHz. The LE system uses 40 RF channels. These RF channels have center frequencies $2402 + k \times 2 \text{ MHz}$, where $k = 0, \dots, 39$.

Regulatory Range	RF Channels
2.400-2.4835 GHz	$f = 2402 + k \times 2 \text{ MHz}$, $k = 0, \dots, 39$

Table 2.1: Operating frequency bands

Fig. 1: Part of the Bluetooth Core Specification Version 5.2

Volume 6, Part A, Section 2. Its applicable extract is shown in Fig. 1. I have no idea where you get 60MHz from!

...

HUMAN HYBRID ELECTRIC VEHICLES

In the 'The Era Of Human Hybrid Electric Vehicles' article published in November 2020 issue, it is mentioned that Twike is a Tesla project. But I don't think so. Can you please confirm?

D. Owens

The author Dr S.S. Verma replies:

You are right. Thanks! In the article, Twike (human-electric hybrid vehicle) is mentioned as a Tesla company project by mistake. Actually, in 1998, FINE Mobile GmbH launched its German Twike production line and finally became the exclusive producer of the

The author Somnath Bera replies:

There is minor flicker with every second but not a blink. The flicker is (generally) not visible at all from a distance of two to three metres. Please check the stability of voltage on the board. Use a good-quality 5V supply (1-amp or more). Put some 470µF or 500µF capacitors across the voltage bus to remove the flicker.

Once you have all the basic data like dd.mm.yy, hh.min.ss, day, and day-of-week (DOW), automatic DST can be achieved. Use one more 'if' statement on top of the time corrections where I used 5 hours 30 minutes for IST corrections. For finding out first Sunday of the month, when (dd > 0 and dd0 and dd ≤ 7 and mm = 10 and DOW = 0) that is the first Sunday of October; put hh = hh + 11. After this, the normal hh corrections will be placed.

The above logic may be placed suitably in two 'if' statements and your